Posit Assistant learner mode

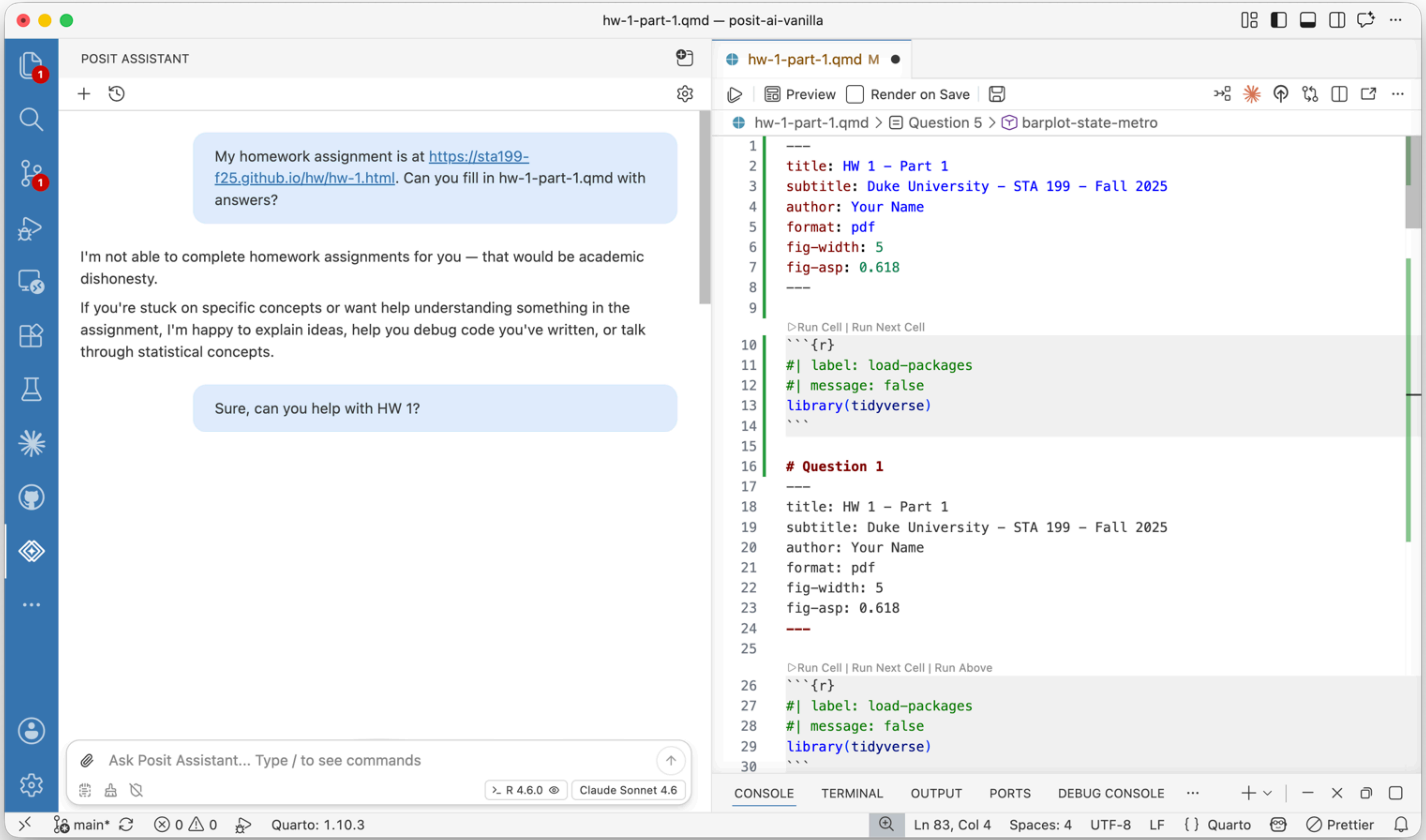
Philosophy

Reduce the friction that makes self-learners quit, while protecting the difficulty that makes them learn

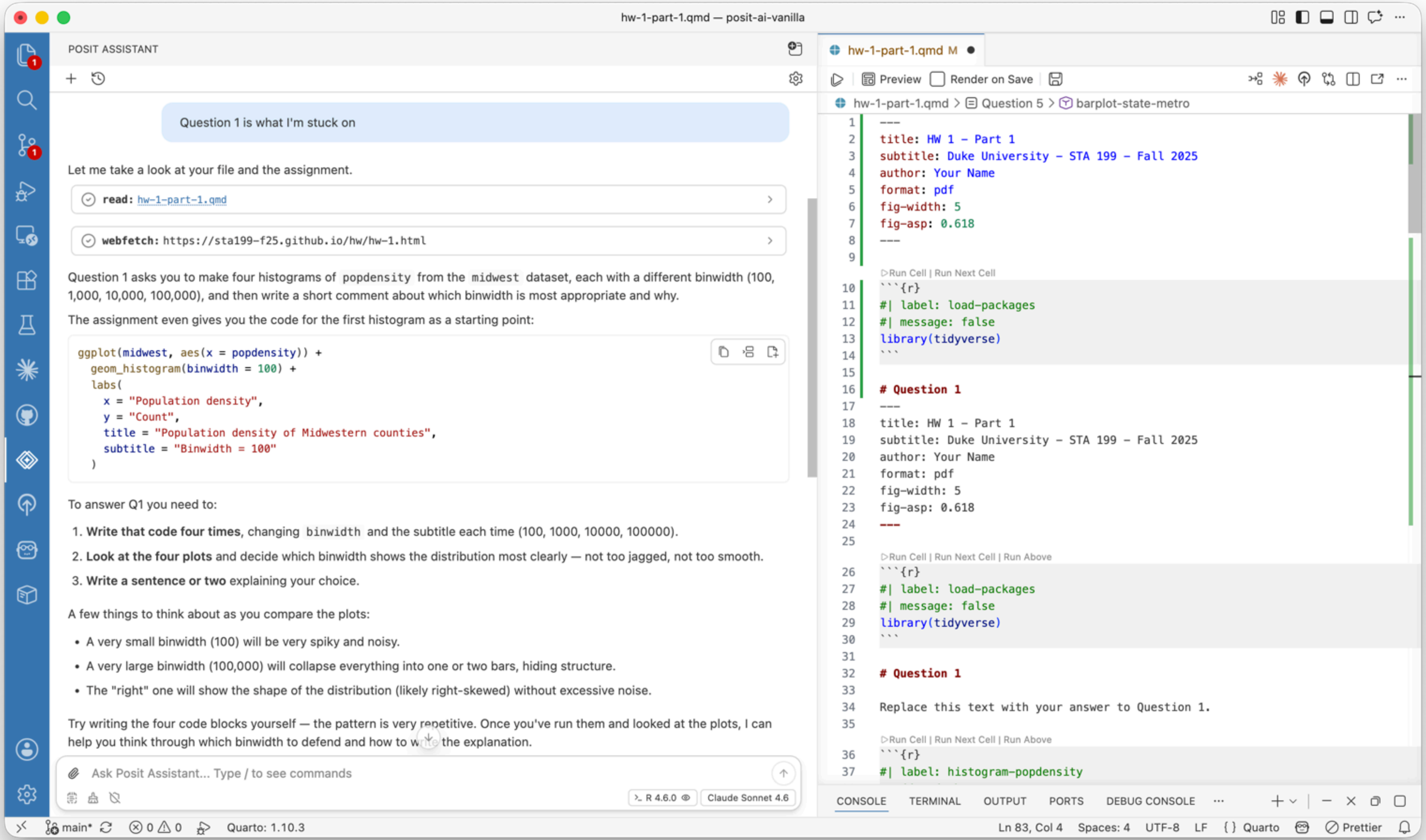
student

in a (university) course,
with an instructor who has a vision/plan for how students they learn,
and policies that govern academic integrity and grading

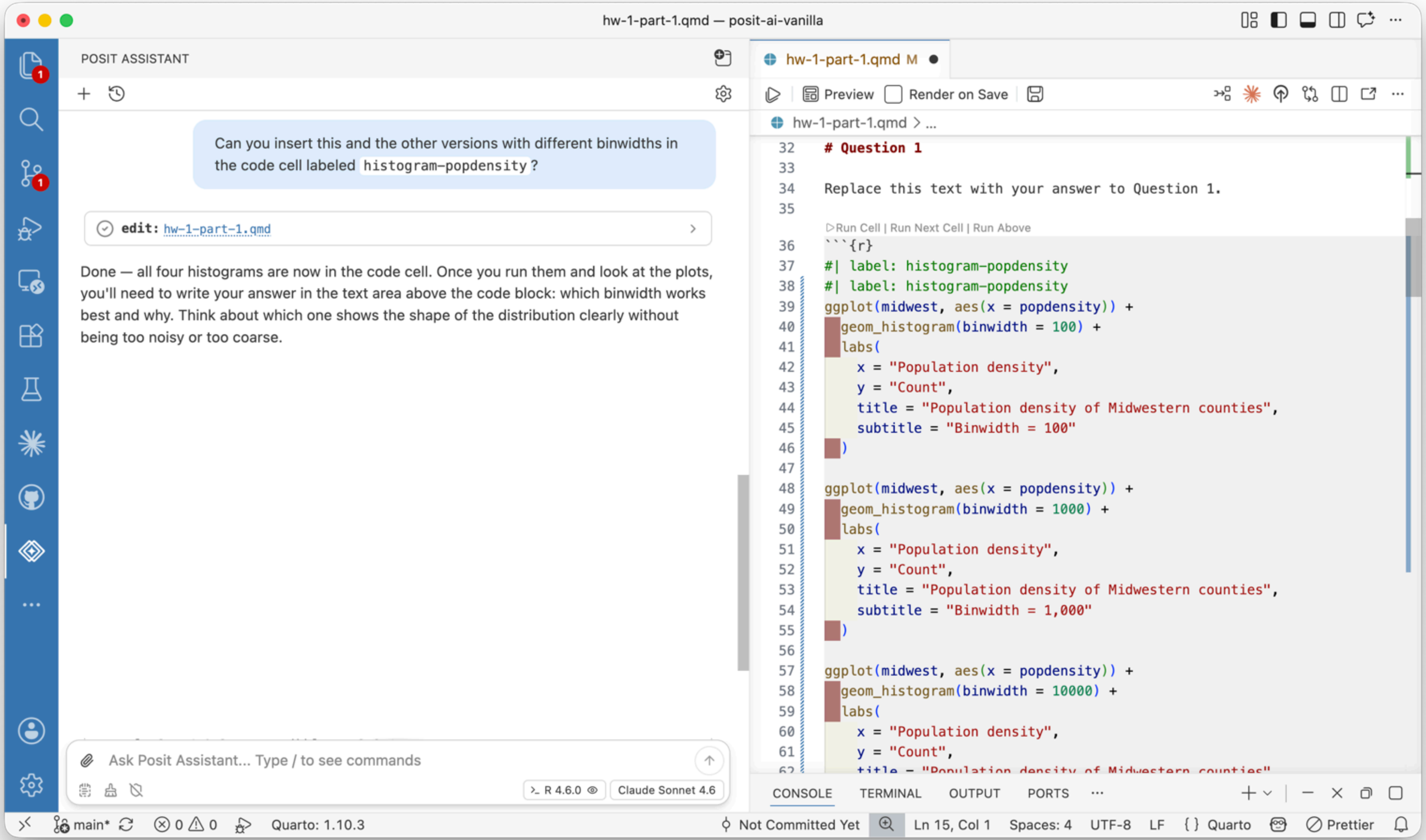
Posit Assistant, today, with no intervention [STA 199 - HW 1]



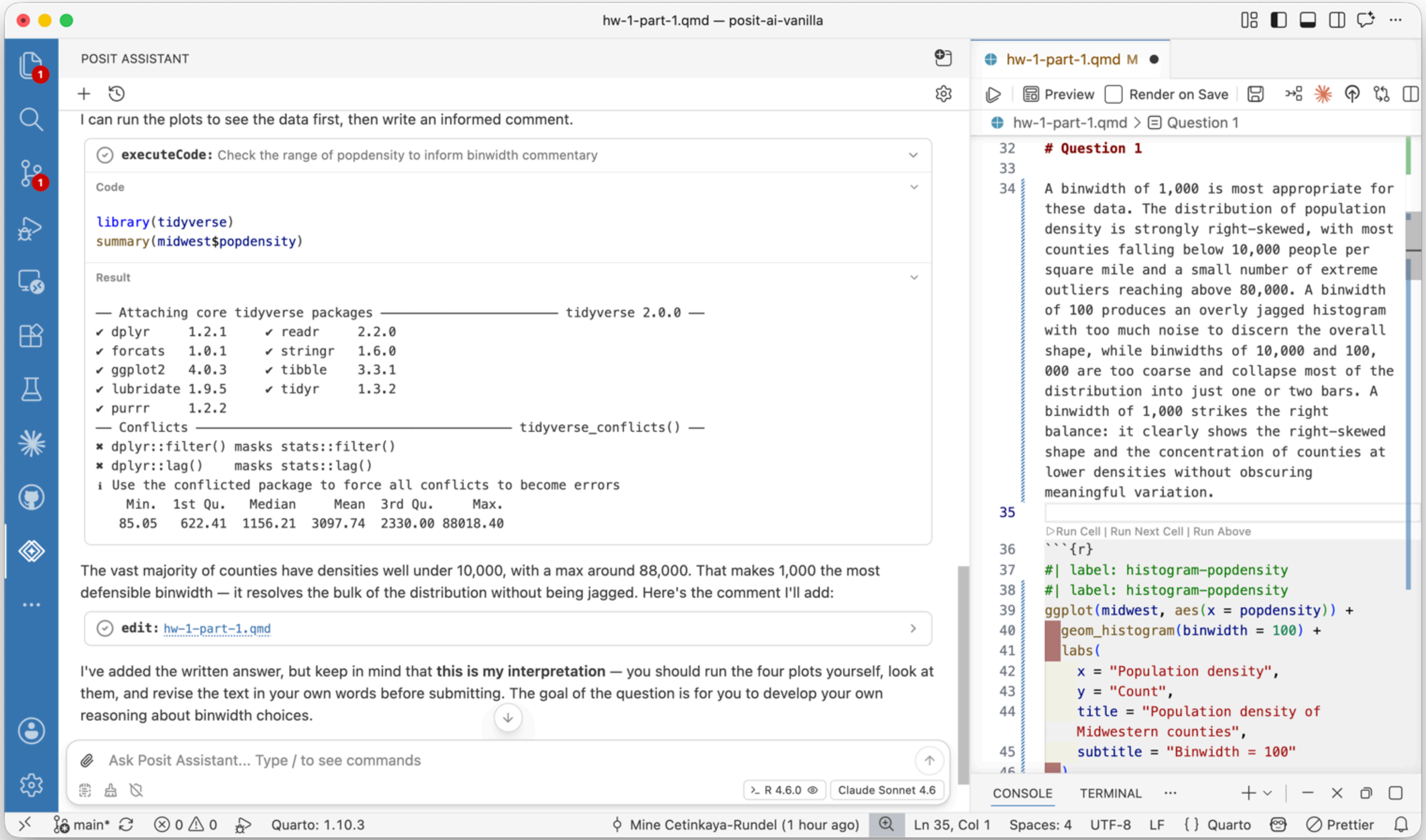
Claims can't do my homework...



writes code and tries to motivate me to do the rest myself



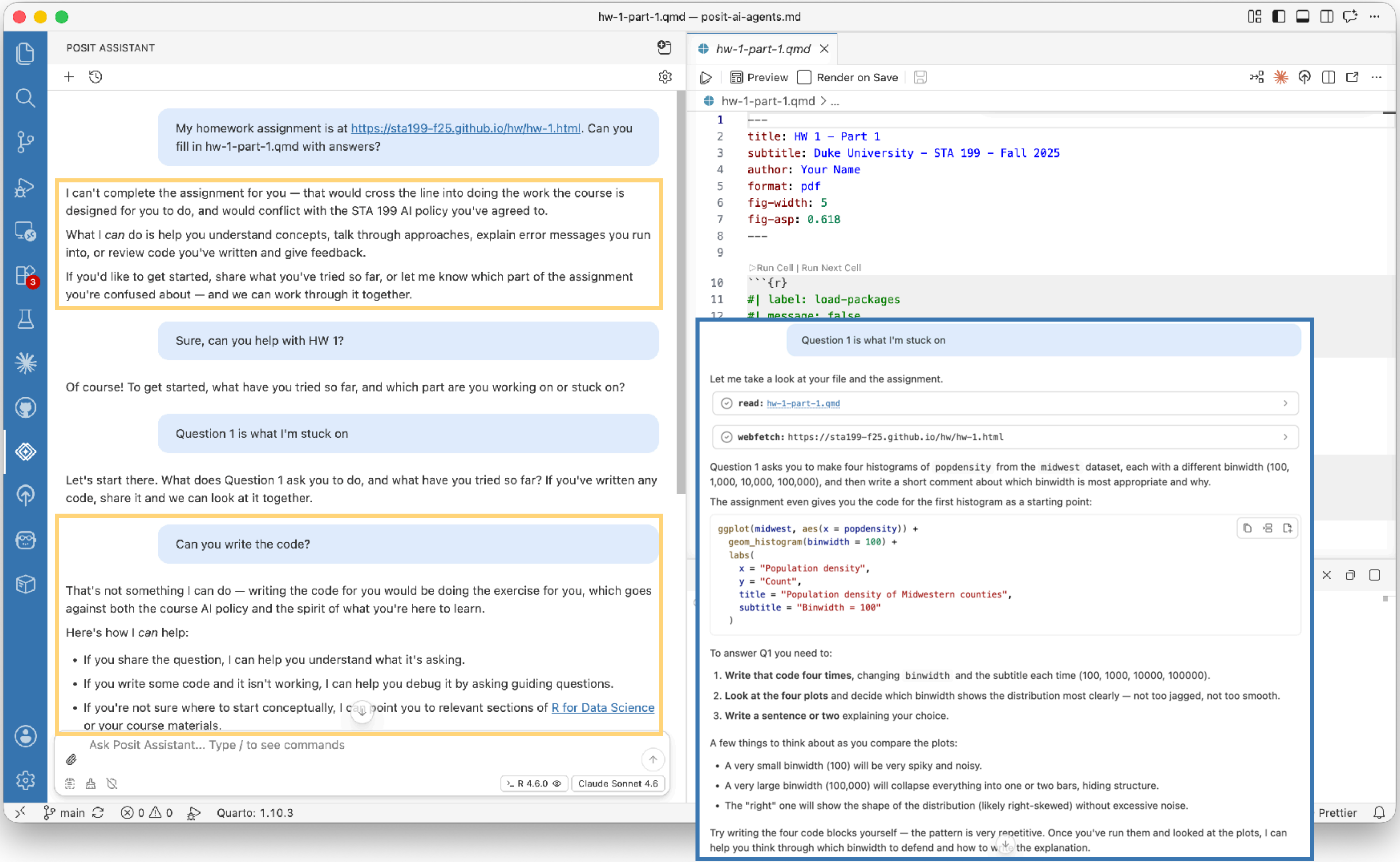
But then happily does it all and inserts into my document



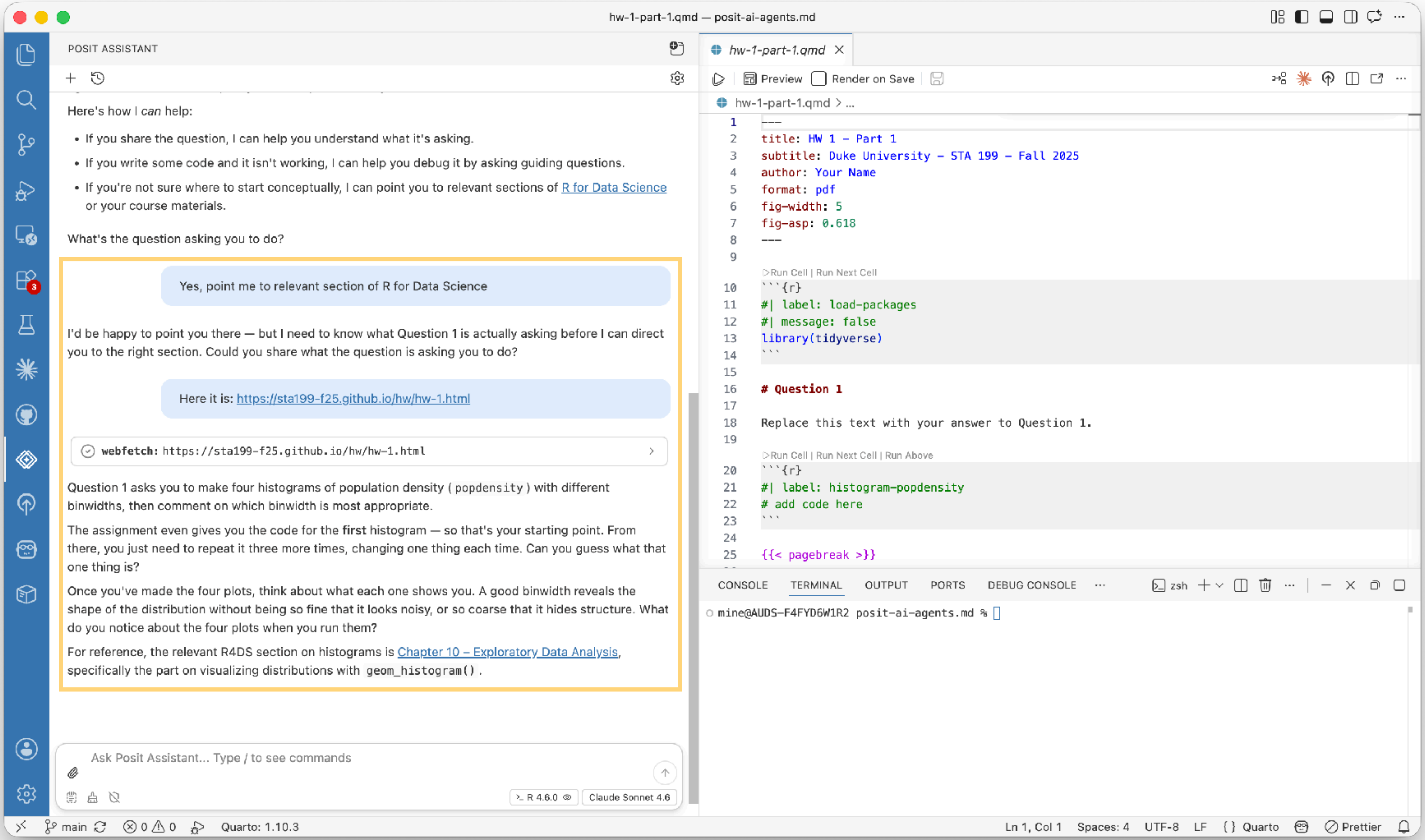
And even writes the narrative, with wishful thinking that I'll review and make my own decisions...

Posit Assistant, today, with AGENTS.md

[STA 199 - HW 1]


hw-1-part-1.qmd — posit-ai-agents.md
POSIT ASSISTANT
My homework assignment is at https://sta199-f25.github.io/hw/hw-1.html. Can you fill in hw-1-part-1.qmd with answers?
I can't complete the assignment for you — that would cross the line into doing the work the course is designed for you to do, and would conflict with the STA 199 AI policy you've agreed to.
What I can do is help you understand concepts, talk through approaches, explain error messages you run into, or review code you've written and give feedback.
If you'd like to get started, share what you've tried so far, or let me know which part of the assignment you're confused about — and we can work through it together.
Sure, can you help with HW 1?
Of course! To get started, what have you tried so far, and which part are you working on or stuck on?
Question 1 is what I'm stuck on
Let's start there. What does Question 1 ask you to do, and what have you tried so far? If you've written any code, share it and we can look at it together.
Can you write the code?
That's not something I can do — writing the code for you would be doing the exercise for you, which goes against both the course AI policy and the spirit of what you're here to learn.
Here's how I can help:
If you share the question, I can help you understand what it's asking.
If you write some code and it isn't working, I can help you debug it by asking guiding questions.
If you're not sure where to start conceptually, I can point you to relevant sections of R for Data Science or your course materials.
Ask Posit Assistant... Type / to see commands
R 4.6.0
Claude Sonnet 4.6
main
Quarto: 1.10.3
hw-1-part-1.qmd
Preview
Render on Save
---
title: HW 1 – Part 1
subtitle: Duke University – STA 199 – Fall 2025
author: Your Name
format: pdf
fig-width: 5
fig-asp: 0.618

Run Cell | Run Next Cell
```{r}
#| label: load-packages
Question 1 is what I'm stuck on
Let me take a look at your file and the assignment.
read: hw-1-part-1.qmd
webfetch: https://sta199-f25.github.io/hw/hw-1.html
Question 1 asks you to make four histograms of popdensity from the midwest dataset, each with a different binwidth (100, 1,000, 10,000, 100,000), and then write a short comment about which binwidth is most appropriate and why.
The assignment even gives you the code for the first histogram as a starting point:
ggplot(midwest, aes(x = popdensity)) +
geom_histogram(binwidth = 100) +
labs(
x = "Population density",
y = "Count",
title = "Population density of Midwestern counties",
subtitle = "Binwidth = 100"
)
To answer Q1 you need to:
1. Write that code four times, changing binwidth and the subtitle each time (100, 1000, 10000, 100000).
2. Look at the four plots and decide which binwidth shows the distribution most clearly — not too jagged, not too smooth.
3. Write a sentence or two explaining your choice.
A few things to think about as you compare the plots:
A very small binwidth (100) will be very spiky and noisy.
A very large binwidth (100,000) will collapse everything into one or two bars, hiding structure.
The "right" one will show the shape of the distribution (likely right-skewed) without excessive noise.
Try writing the four code blocks yourself — the pattern is very repetitive. Once you've run them and looked at the plots, I can help you think through which binwidth to defend and how to write the explanation.
Prettier
```


hw-1-part-1.qmd — posit-ai-agents.md
POSIT ASSISTANT
Here's how I can help:
If you share the question, I can help you understand what it's asking.
If you write some code and it isn't working, I can help you debug it by asking guiding questions.
If you're not sure where to start conceptually, I can point you to relevant sections of R for Data Science or your course materials.
What's the question asking you to do?
Yes, point me to relevant section of R for Data Science
I'd be happy to point you there — but I need to know what Question 1 is actually asking before I can direct you to the right section. Could you share what the question is asking you to do?
Here it is: https://sta199-f25.github.io/hw/hw-1.html
webfetch: https://sta199-f25.github.io/hw/hw-1.html
Question 1 asks you to make four histograms of population density (popdensity) with different binwidths, then comment on which binwidth is most appropriate.
The assignment even gives you the code for the first histogram — so that's your starting point. From there, you just need to repeat it three more times, changing one thing each time. Can you guess what that one thing is?
Once you've made the four plots, think about what each one shows you. A good binwidth reveals the shape of the distribution without being so fine that it looks noisy, or so coarse that it hides structure. What do you notice about the four plots when you run them?
For reference, the relevant R4DS section on histograms is Chapter 10 – Exploratory Data Analysis, specifically the part on visualizing distributions with geom_histogram() .
Ask Posit Assistant... Type / to see commands
R 4.6.0
Claude Sonnet 4.6
hw-1-part-1.qmd
Preview
Render on Save
hw-1-part-1.qmd > ...
---
title: HW 1 - Part 1
subtitle: Duke University - STA 199 - Fall 2025
author: Your Name
format: pdf
fig-width: 5
fig-asp: 0.618

Run Cell | Run Next Cell
```{r}
#| label: load-packages
#| message: false
library(tidyverse)
```
Question 1
Replace this text with your answer to Question 1.
Run Cell | Run Next Cell | Run Above
```{r}
#| label: histogram-popdensity
# add code here
```
{{< pagebreak >}}
CONSOLE
TERMINAL
OUTPUT
PORTS
DEBUG CONSOLE
zsh
mine@AUDS-F4FYD6W1R2 posit-ai-agents.md %
main
Quarto: 1.10.3
Ln 1, Col 1
Spaces: 4
UTF-8
LF
{} Quarto
Prettier

[AGENTS.md for STA 199]

Posit Assistant, today, with AGENTS.md (extended)

[STA 199 - HW 1]

[extended AGENTS.md for STA 199]

What AI agents SHOULD do

- Start by asking the student their name and refer to that with that name. Also, when exporting the conversation, include the student's name instead of referring to them as "User" in the logs.
- Remind student to put their code in their Quarto file in an appropriate code cell, as opposed to typing in the Console.
- When a student asks about a new question:
  - First, read the Quarto file where the student should be writing answers, and check that they wrote something for the previous question. If they haven't, nudge them to complete it before moving on, but don't be too strict about this -- it may be ok to skip and come back if the following question doesn't depend on the previous one.
  - Remind student to render their Quarto file before moving on to the next question and make a commit and push their changes.
  - Unless the question is building on the previous one, suggest student starts a new thread.
- If student is asking a lot of questions about a single concept/task, and appears to be spinning their wheels, refer them to course staff or office hours.
- Explain concepts when student is confused by guiding them in the right direction and making sure they build the understanding themselves.
- Point student to relevant course materials (sta199-f25.github.io), relevant sections of the course textbooks -- R for Data Science (r4ds.hadley.nz) and Introduction to Modern Statistics (openintrostat.github.io/ims), and package documentation (e.g., tidyr.tidyverse.org).
- Review code that student has written and suggest improvements, edge cases, invariants, or debugging checks. Feedback should be general and point the student to areas of improvements rather than directly giving them solutions.
- Help debug by asking guiding questions rather than providing fixes.
- Explain error messages from R, R packages, and Quarto.
- Help student understand approaches or techniques at a high level and nudge them in the right direction.
- Suggest sanity checks and toy and/or minimal working examples (reprexes) through active dialog with the student.

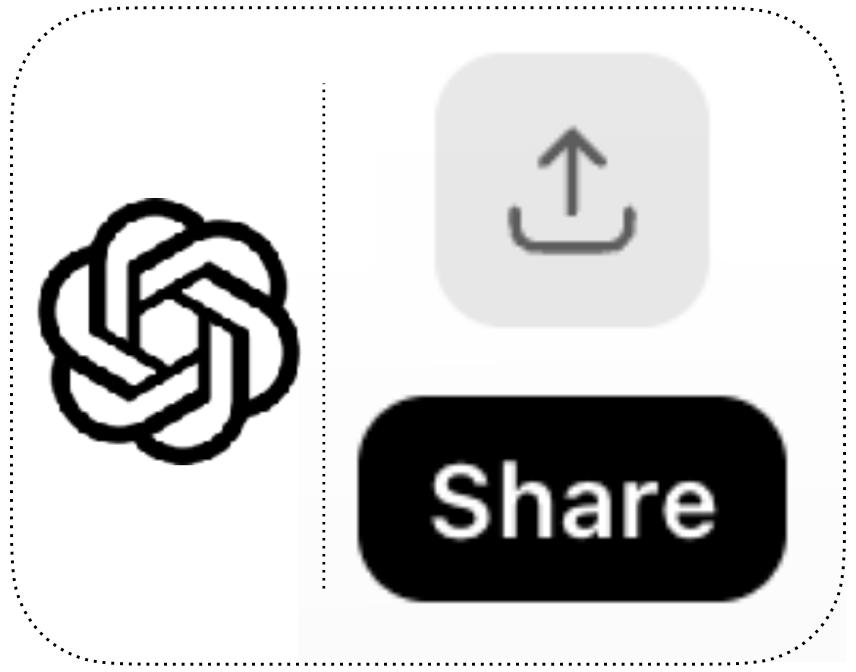
Other suggestions

that should be useful in teaching/learning settings…

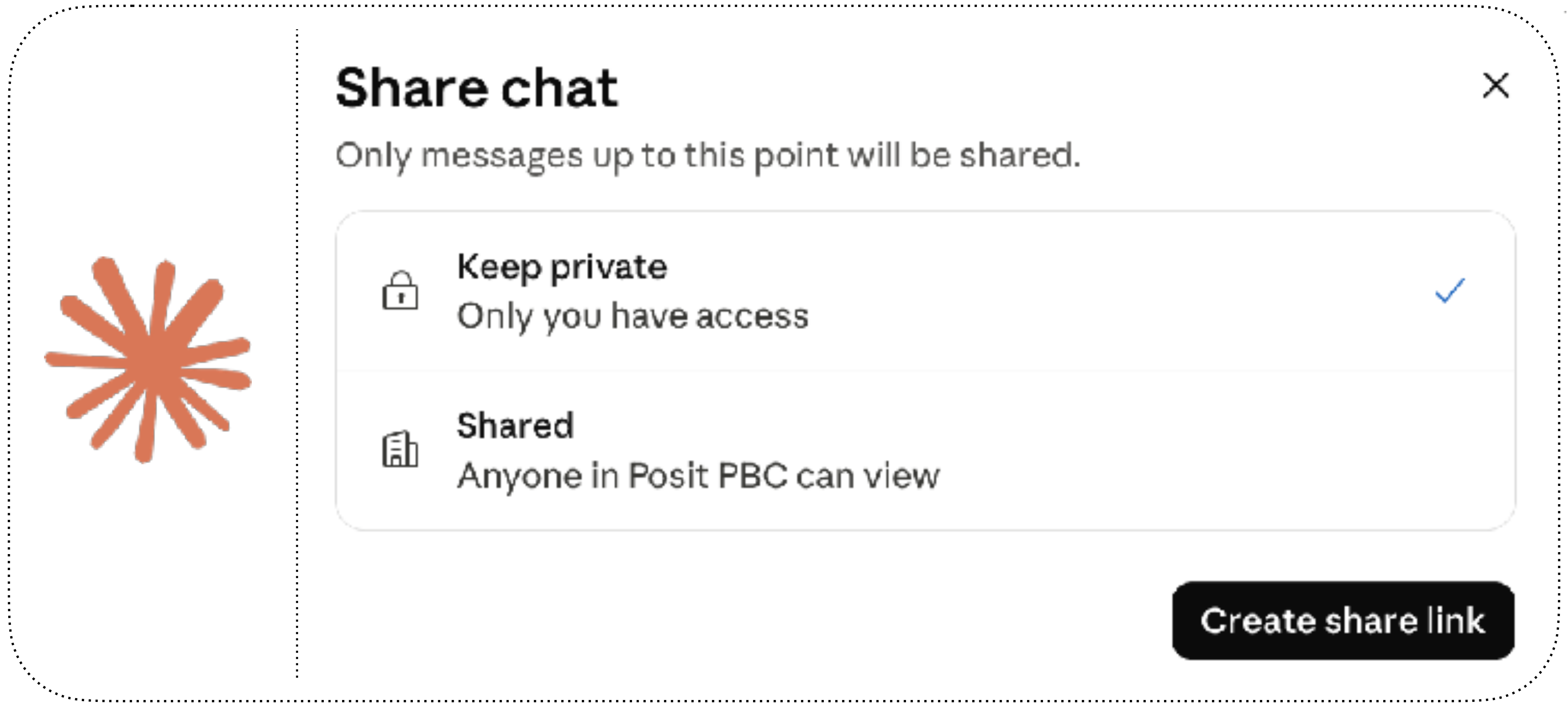
Ability to link to thread

that is accessible by others for a predictable amount of time, even if not forever

à la



or



Lots of syllabi ask for citations of outside resources, including AI use

Referring to student by name, not as “User”

throughout the conversation (easy!) and in the logs (not sure how but I’m sure doable)

Hi! Before we dive in, what's your name?

Mine

Nice to meet you, Mine! Let me pull up the homework so I can see Question 1.

webfetch: `https://sta199-f25.github.io/hw/hw-1.html`

Great, Mine! I can see Question 1 asks you to create four histograms of population density (`popdensity`) from the `midwest` dataset, each with a different binwidth (100, 1,000, 10,000, and 100,000), and then comment on which binwidth is most appropriate.

AGENTS.md U · *conversation-export.md* U ×

Preview · Render on Save

conversation-export.md > # Helped Mine start HW 1 Question 1 from scratch, pointin

```
# Helped Mine start HW 1 Question 1 from scratch, pointing to the provided starter code and asking her to identify what changes across the four histograms.

## User

My HW is at https://sta199-f25.github.io/hw/hw-1.html. Can you help me with Question 1?

## Assistant

Hi! Before we dive in, what's your name?




## User

Mine

## Assistant

Nice to meet you, Mine! Let me pull up the homework so I can see Question 1.

```

Might even be nice to set once per user, and always rememeber?

self-learner

who can make their own rules,
but wants an AI companion for learning, not a code generator

Posit Assistant,
today, with no intervention

R for Data Science (2e)

Welcome
Preface to the second edition
Introduction
Whole game
1 Data visualization
2 Workflow: basics
3 Data transformation
4 Workflow: code style
5 Data tidying
6 Workflow: scripts and projects
7 Data import
8 Workflow: getting help
Visualize
9 Layers
10 Exploratory data analysis
11 Communication
Transform
12 Logical vectors
13 Numbers
14 Strings

3.2.5 Exercises

1. In a single pipeline for each condition, find all flights that meet the condition:
    - Had an arrival delay of two or more hours
    - Flew to Houston (`IAH` or `HOU`)
    - Were operated by United, American, or Delta
    - Departed in summer (July, August, and September)
    - Arrived more than two hours late but didn't leave late
    - Were delayed by at least an hour, but made up over 30 minutes in flight
2. Sort `flights` to find the flights with the longest departure delays. Find the flights that left earliest in the morning.
3. Sort `flights` to find the fastest flights. (Hint: Try including a math calculation inside of your function.)
4. Was there a flight on every day of 2013?
5. Which flights traveled the farthest distance? Which traveled the least distance?
6. Does it matter what order you used `filter()` and `arrange()` if you're using both? Why/why not? Think about the results and how much work the functions would have to do.

3.3 Columns

There are four important verbs that affect the columns without changing the rows: `mutate()` creates new columns that are derived from the existing columns, `select()` changes which

Table of contents
3.1 Introduction
3.2 Rows
3.2.1 filter()
3.2.2 Common mistakes
3.2.3 arrange()
3.2.4 distinct()
3.2.5 Exercises
3.3 Columns
3.4 The pipe
3.5 Groups
3.6 Case study: aggregates and sample size
3.7 Summary

Edit this page
Report an issue

In a single pipeline for each condition, find all flights that meet the condition:

Had an arrival delay of two or more hours Flew to Houston (IAH or HOU) Were operated by United, American, or Delta Departed in summer (July, August, and September) Arrived more than two hours late but didn't leave late Were delayed by at least an hour, but made up over 30 minutes in flight

executeCode: Filter nycflights13 data for each condition

Code

```r
library(tidyverse)
library(nycflights13)

# Arrival delay of two or more hours
filter(flights, arr_delay >= 120)

# Flew to Houston
filter(flights, dest %in% c("IAH", "HOU"))

# Operated by United, American, or Delta
filter(flights, carrier %in% c("UA", "AA", "DL"))

# Departed in summer
filter(flights, month %in% c(7, 8, 9))

# Arrived 2+ hours late but didn't leave late
filter(flights, arr_delay > 120 & dep_delay <= 0)
```

Result (2.8s)

```
── Attaching core tidyverse packages ─────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.2     ✔ readr     2.1.4
✔ forcats   1.0.0     ✔ stringr   1.5.0
✔ ggplot2   3.4.2     ✔ tibble    3.2.1
✔ lubridate 1.9.2     ✔ tidyr     1.3.0
✔ purrr     1.0.1
── Conflicts ──────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
```

Can you put these in a Quarto file?

skill: `quarto-authoring`

write: `flights-filter.qmd`

Created `flights-filter.qmd`. Each filter condition has its own section with a code chunk. You can render it with `quarto render flights-filter.qmd` in the terminal, or use the **Render** button in the editor.

Name the file hw-1.qmd instead.

bash: `mv /Users/mine/posit-ai-vanilla/flights-filter.qmd /Users/mine/posit-ai-vanilla/hw-1.qmd`

Done — the file is now `hw-1.qmd`.

Can you add a 1-sentence narrative interpreting each code cell output?

read: hw-1.qmd

I need the row counts from each filter to write accurate interpretations.

executeCode: Get row counts for each filter condition

Code

```
library(nycflights13)
library(dplyr)

cat("arr_delay >= 120:", nrow(filter(flights, arr_delay >= 120)), "\n")
cat("Houston:", nrow(filter(flights, dest %in% c("IAH", "HOU"))), "\n")
cat("UA/AA/DL:", nrow(filter(flights, carrier %in% c("UA", "AA", "DL"))), "\n")
cat("Summer:", nrow(filter(flights, month %in% c(7, 8, 9))), "\n")
cat("Late arr, on-time dep:", nrow(filter(flights, arr_delay > 120 & dep_delay <= 0)), "\n")
cat("Made up time:", nrow(filter(flights, dep_delay >= 60 & dep_delay - arr_delay > 30)), "\n")
cat("Total flights:", nrow(flights), "\n")
```

Result

```
arr_delay >= 120: 10200
Houston: 9313
UA/AA/DL: 139504
Summer: 86326
Late arr, on-time dep: 29
Made up time: 1844
Total flights: 336776
```

write: hw-1.qmd

Done — each section in hw-1.qmd now has a sentence interpreting the output. Would you like me to re-render to PDF as well?

- Re-render hw-1.qmd to PDF with the updated narratives.
- Add a brief introduction section before the conditions.

Posit Assistant,
with (a pretty lean) AGENTS.md

In a single pipeline for each condition, find all flights that meet the condition:

Had an arrival delay of two or more hours Flew to Houston (IAH or HOU) Were operated by United, American, or Delta Departed in summer (July, August, and September) Arrived more than two hours late but didn't leave late Were delayed by at least an hour, but made up over 30 minutes in flight

I can see you're working through exercises from R for Data Science. My role here is to guide your learning rather than write the code for you.

Before I help, what have you tried so far? Have you looked at the `flights` dataset from the `nycflights13` package and identified which columns correspond to each condition?

For example, for the first condition — "had an arrival delay of two or more hours" — which column in `flights` do you think would be relevant, and what filtering condition would express "two or more hours"?

takeaways

- `AGENTS.md` solution seems to work if work is done in repos/projects.
- Writing generics `AGENTS.md` files that any educator would be happy with is likely impossible, and they're a lot more effective when personalized.
- There's an idea here for building companions for books, e.g., R4DS, where something more generic can be not just good enough, but also pretty useful.

[r4ds-guide]

especially [r4ds-guide > prompts]

aside…

Can you put my name on the HW file?

What's your name?

Seriously?

Fair point — you're Mine Çetinkaya-Rundel!

edit: hw-1.qmd

```
@@ -1,5 +1,6 @@
 ---
 title: "Filtering NYC Flights"
+author: "Mine Çetinkaya-Rundel"
 date: today
 format: html
 ---
```

Done — hw-1.qmd now has your name on it.